

# Conditions that Guarantee Perpendicularity

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**Perpendicular lines** are lines that intersect and form a right angle ( $90^\circ$ ).

# Perpendicular Slopes

In the coordinate plane, two non-vertical lines are perpendicular if their **slopes** are negative reciprocals of each other.

# Slope Condition

If the slope of one line is  $m_1$  and the slope of the other line is  $m_2$ , then the lines are perpendicular when

$$m_1 m_2 = -1.$$

# Negative Reciprocal

The **negative reciprocal** of a number is obtained by reversing the numerator and denominator and changing the sign.

For example,

$$\frac{2}{3} \longrightarrow -\frac{3}{2}.$$

# Horizontal and Vertical Lines

A **horizontal line** and a **vertical line** are always perpendicular.

$$y = c \quad \text{and} \quad x = c$$

# How to Determine Perpendicularity

To determine whether two lines are perpendicular:

- Find the slope of each line.

- Check whether the slopes are negative reciprocals.

- Alternatively, verify that the product of the slopes is  $-1$ .

# Examples

## Example 1

Determine whether the lines with slopes 2 and  $-\frac{1}{2}$  are perpendicular.

$$2\left(-\frac{1}{2}\right) = -1$$

Therefore, the lines are perpendicular.

## Example 2

Determine whether the lines with slopes 3 and  $-\frac{1}{2}$  are perpendicular.

$$3 \left( -\frac{1}{2} \right) = -\frac{3}{2} \neq -1$$

Therefore, the lines are not perpendicular.

# Example 3

Find the value of  $a$  so that the lines

$$y = ax + 7$$

and

$$y = -\frac{1}{4}x + 2$$

are perpendicular.

# Example 3

The slopes are  $a$  and  $-\frac{1}{4}$ .

$$a \left( -\frac{1}{4} \right) = -1$$

$$a = 4$$

Therefore,

$$\boxed{a = 4}$$

## Example 4

Determine whether the line passing through  $(-2, 1)$  and  $(4, -2)$  is perpendicular to the line whose slope is 2.

# Example 4

Find the first slope:

$$m = \frac{-2 - 1}{4 - (-2)} = \frac{-3}{6} = -\frac{1}{2}.$$

Then,

$$\left(-\frac{1}{2}\right)(2) = -1.$$

Therefore, the lines are perpendicular.

# Practice Exercises

# Practice 1

Determine whether the lines with slopes  $-\frac{4}{5}$  and  $\frac{5}{4}$  are perpendicular.

# Practice 2

Determine whether the lines

$$y = -3x + 5$$

and

$$y = \frac{1}{3}x - 2$$

are perpendicular.

# Practice 3

Find the value of  $a$  so that the lines

$$y = ax + 7$$

and

$$y = -\frac{1}{4}x + 2$$

are perpendicular.

## Practice 4

Determine whether the line passing through  $(-2, 1)$  and  $(4, -2)$  is perpendicular to the line whose slope is 2.

## Practice 5

Show that the lines passing through  $(0, 0)$  and  $(2, 6)$ , and through  $(1, 4)$  and  $(4, 3)$  are perpendicular.

# Activities

# Activity 1

Determine whether the lines with slopes 4 and  $-\frac{1}{4}$  are perpendicular.

# Activity 2

Determine whether the lines

$$y = 5x - 2$$

and

$$y = -\frac{1}{5}x + 8$$

are perpendicular.

## Activity 3

Find the value of  $k$  so that the lines with slopes  $k$  and  $\frac{2}{7}$  are perpendicular.

## Activity 4

Determine whether the line passing through  $(2, 1)$  and  $(8, 13)$  is perpendicular to the line whose slope is  $-\frac{1}{2}$ .

# Activity 5

Determine whether the horizontal line

$$y = -4$$

and the vertical line

$$x = 7$$

are perpendicular.

Thank you!